System Startup Guide

System Initialization

1

Switch ON the solar MCB, battery MCCB, at inverter power switch (with cooldown). Then activate the grid breaker, HelioGrid device, sensors, at Arduino. Proper sequence ensures safe hybrid power startup.

Real-Time Monitoring

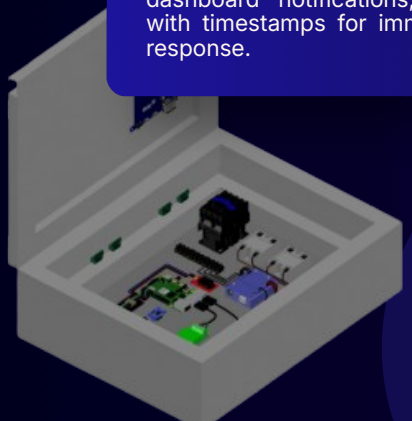
2

HelioGrid continuously tracks voltage flow from grid, solar, and battery sources. Data is processed by rule-based logic to detect anomalies in real time via PZEM sensors and Arduino input.

Anomaly Detection & Alerts

3

The system classifies abnormal voltage into spike, drift, and dropout. Alerts are triggered via buzzer, dashboard notifications, and logged with timestamps for immediate admin response.



Technical Specifications

CORE CONTROLLER

- Raspberry Pi 4B - GUI, rule-based logic & database management.
- Arduino Uno - Sensor input & hardware I/O.

POWER & SWITCHING

- ONE-SOLAR Hybrid Inverter (3kW) - MPPT solar charging, battery charging, AC output.
- SSR K1/K2 + CHNT Contactor - relay switching between inverter and grid.

SENSORS & I/O

- PZEM-004T - grid & inverter AC, Solar & battery DC monitoring.
- Voltage Divider + INA226 - battery SOC/current monitoring
- DHT22 - enclosure temperature/humidity
- LCD Touchscreen (5"x7") - local dashboard
- Buzzer - audible alarm for critical anomalies

SOFTWARE STACK

- JavaScript – core scripting for frontend logic
- React – component-based UI framework for web dashboard
- TypeScript – typed superset of JavaScript for safer code
- HTML & CSS – structure and styling of web interface
- Flask – Python micro-framework for backend API
- Python – anomaly detection logic and rule-based classification
- MySQL – relational database for logging anomalies and transactions

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HELIOGRID

BSCPE THESIS · 2026

Hybrid Power Anomaly Monitor

Hybrid Power Integration & Rule-Based Anomaly Detection — a fully automated solution for solar-grid-battery monitoring, anomaly classification, and real-time campus resilience.

WHERE DETECTION MEETS AUTOMATION



83.93%	Dropout Anomalies
9.12%	Spike Anomalies
6.95%	Drift Anomalies
300+	Total logged

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The Need for HelioGrid

Manual Monitoring Fails

Traditional hybrid setups lack automated anomaly detection, leading to unnoticed spikes, drifts, and dropouts that damage equipment and reduce energy efficiency.

No Real-Time Audit Trail

Without automated logging, anomalies go unrecorded. Preventive maintenance and accountability are difficult to enforce.

Efficiency Demand

Solar integration is growing in Philippine HEIs. HelioGrid delivers a localized, cost-effective hybrid monitoring solution to fill this gap.

OVERALL ANOMLY DETECTION RESULT

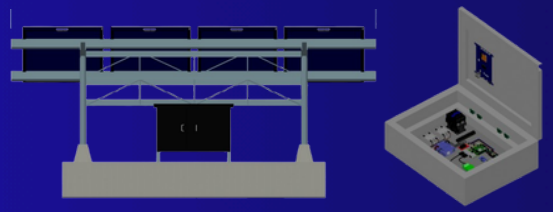
DROPOUT	SPIKE	DRIFT
28,291	1,957	3,460

The values shown above represent the total anomalies detected during data gathering under both no-load and with-load conditions. Each voltage/current reading from grid, solar, and battery sources was classified into dropout, spike, or drift using the rule-based detection engine. The counts (28,291 Dropout, 3,460 Spike, 1,957 Drift) therefore account for all 33,708 anomalies recorded in the dataset.

Project Objectives

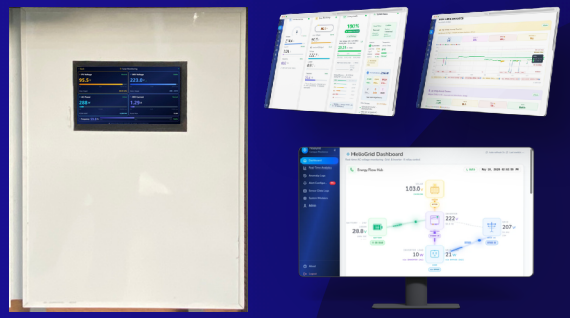
1. Automate anomaly detection in hybrid power systems
2. Implement rule-based classification for spike, drift, and dropout
3. Log anomalies in real time with admin dashboard access
4. Evaluate performance across accuracy, precision, and usability

SYSTEM ARCHITECTURE



The HelioGrid System Architecture integrates grid, solar, and battery sources into a single control unit. Key hardware includes breakers, inverter, Arduino with PZEM sensors, and a Raspberry Pi. Sensor data flows through the Arduino to the Raspberry Pi, stored in a database, and displayed on the dashboard. A rule-based detection layer classifies anomalies as dropout, spike, or drift, while buzzer and email alerts ensure safety. An external kiosk rack provides accessible monitoring outside the control room.

WEB APPLICATION | KIOKS SYSTEM



The HelioGrid Web Application & Kiosk System complements the hardware with a real-time admin dashboard. Voltage and current readings are visualized through energy flow, graphs and logs supported by buzzer and email alerts for anomalies. Accessible via a Chromium browser on any WiFi-connected device, the kiosk display allows administrators to monitor and evaluate system performance efficiently.

Significance of Study

- *Campus resilience – HelioGrid strengthens hybrid power systems by preventing interruptions through automated anomaly detection.*
- *Preventive maintenance – Real-time logging of anomalies enables faster corrective action and reduces equipment damage.*
- *Localized integration – Provides a cost-effective monitoring solution tailored for Philippine HEIs adopting solar hybrid setups.*
- *Research contribution – Supports sustainable energy monitoring studies and advances rule-based anomaly detection methods.*

Future Work

- *AI prediction – Integrating machine learning models for adaptive anomaly forecasting.*
- *Mobile app – Developing a portable interface for real-time monitoring and alerts.*
- *Cloud dashboard – Expanding to remote access and centralized data visualization.*
- *Scalability – Extending HelioGrid to larger hybrid systems and multi-campus deployments.*

Performance Evaluation

97.71%	Overall Accuracy
90.01%	Precision
92.32%	Recall
91.15%	F-1 Score

